Public vs. Private Networks — Guided Notes ANSWER KEY

Unit 1.4.1 | CompTIA Network+ N10-009 Obj. 1.4 | N10-008 1.4

For instructor use / distribute after students complete the notes.

Question 1

List the three RFC 1918 private IP ranges and their CIDR notation:

1. _______ (_______)
2. _______ (_______)
3. _______ (_______)

Answer: 1. 10.0.0.0 – 10.255.255.255 (10.0.0.0/8); 2. 172.16.0.0 – 172.31.255.255 (172.16.0.0/12); 3. 192.168.0.0 – 192.168.255.255 (192.168.0.0/16).

Question 2

Private IP addresses are never _______ on the public internet. The reason private addressing exists is that IPv4 only provides approximately _______ total addresses, which is not enough for every device in the world to have a unique public IP.

Answer: Routed; 4.3 billion.

Question 3

NAT stands for _______. When a device on a private network sends traffic to the internet, the router replaces the _______ IP with its own _______ IP before forwarding the packet.

Answer: Network Address Translation; private source; public.

Question 4

PAT (Port Address Translation) allows an entire network to share a _______ public IP address. It tracks which internal device made which request by assigning unique _______ numbers to each outbound connection.

Answer: Single; source port.

Question 5

APIPA assigns addresses in the _______ range when a device cannot reach a _______. An APIPA address means the device can only communicate with _______ on the same segment and cannot reach _______.

Answer: 169.254.0.0/16; DHCP server; other APIPA devices; any gateway or remote network.

Question 6

When a technician sees a 169.254.x.x address on a workstation, list three things they should immediately check:

1. _______
2. _______
3. _______

Answer: 1. Is the DHCP server running? 2. Is the DHCP scope exhausted (out of addresses)? 3. Is the network cable connected / is there a link? (Any three valid checks.)

Question 7

The loopback address is _______. Pinging this address tests whether _______ is functioning on the local machine. Traffic sent to this address never leaves the device and never touches _______.

Answer: 127.0.0.1; the TCP/IP stack; a cable or NIC.

Question 8 — Real World Application

REAL WORLD: A user calls to say they can't access anything on the network or the internet. You remotely view their screen and see their IP address is 169.254.83.42. Walk through your diagnostic steps in order.

Answer: 1. Confirm APIPA — DHCP failed. 2. Check physical connection (cable seated, link light on). 3. Try to ping the DHCP server by IP from another device to confirm it's reachable. 4. Check if DHCP service is running on the server. 5. Check if the DHCP scope is exhausted. 6. If physical and server are fine, try ipconfig /release and /renew to force a new DHCP request. 7. If still failing, check switch port, VLAN assignment, and DHCP relay/helper address configuration.